

Arunangshu Mojumder Raatul

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EDUCATION

SRM Institute of Science and Technology

Chennai, India

Master of Technology in Information Security and Cyber Forensics

May 2024 – May 2026

- CGPA: **9.62/10.00** (Graduated with First Class Distinction)

University of Liberal Arts Bangladesh

Dhaka, Bangladesh

Bachelor of Science in Computer Science and Engineering

2019 - 2023

- CGPA: **3.86/4.00** (Graduated with Magna cum Laude Distinction)

Technical Skills

Programming: Python, C, C++, JavaScript, SQL, Solidity, Dart, LaTeX

Blockchain & Web3 Tools: Hardhat, Truffle, Ganache, Web3.js, Ethers.js, Clarity

Penetration Testing: Web Application Security Testing, Vulnerability Assessment, OWASP Top 10

Tools: Kali Linux, Burp Suite, Nmap, Wireshark, Metasploit, Recon-ng, Autopsy, Qualys VMDR

Network Security: TCP/IP, DNS, HTTP/HTTPS, Packet Analysis, Firewall Concepts

Blue Team: Threat Hunting, SIEM Fundamentals, Incident Response, OSINT

EXPERIENCE

Junior Research Assistant – Blockchain and Cryptography Project

(March 2024 – August 2024)

University of Liberal Arts Bangladesh

Dhaka, Bangladesh

- Contributed to a Blockchain and Cryptography Project Funded by Ministry of Science and Technology

Lab Assistant (Certified Ethical Hacker)

(June 2024 – July 2024)

IBCS-Primax

Dhaka, Bangladesh

- Lab Assistant for CEH training to Government officials

Teaching Assistant

(February 2023 – April 2024)

University of Liberal Arts Bangladesh

Dhaka, Bangladesh

- Mentored students in AI, Computer Security, and Networking; supported labs and assessments

NOTABLE PROJECTS

Decentralized Identity-Bound Software Licensing Using modified NFTs and Polymorphic Cryptographic Vaults

- Built a decentralized licensing framework using **ERC-5192 Tokens** on Ethereum Sepolia.
- Developed a **polymorphic cryptographic** engine mutating key derivation logic achieving **<1ms overhead**

CyberRecon – Automated Reconnaissance and Security Assessment Framework

- **Modular tool** for **vulnerability scanning** and **security analysis** integrating **Nmap**, **CVE mapping** and **header analysis**.
- Implemented automated **false-positive filtering logic** to refine **scan accuracy** using CVE data.

Blockchain Based Academic Credentials Verifier with Self Made Lightweight Encryption

- Developed a **secure verification system** using Ethereum, JS, and MySQL with timestamp-based encryption.
- Focused on **data confidentiality**, **integrity verification**, and **smart contract validation**.

Certifications

- **Certified Ethical Hacker (CEH)**
February 2025, Issued By: EC-Council
- **Google Cybersecurity Professional Certificate**
July 2025, Issued By: Google
- **Blue Team Junior Analyst**
July 2025, Issued By: Security Blue Team
- **ISO/IEC 27001:2022 Lead Auditor**
July 2025, Issued By: Mastermind Assurance
- **ISO/IEC 42001:2023 Lead Auditor**
June 2025, Issued By: Mastermind Assurance

- **Vulnerability Management Detection and Response**

June 2025, Issued By: Qualys

English Proficiency

- **IELTS: 8.0** (Listening: 8.5, Writing: 7.5, Reading: 7.5, Speaking: 8.0)

Publications (Peer Reviewed)

1. Ephemeral Cryptographic Architecture for Blockchain-Based Software Licensing – [IEEE Xplore](#)
2. RMT: Lightweight Encryption Algorithm for Blockchain Certificate Verification — [IEEE Xplore](#)
3. Verifi-chain: Credential Verifier using Blockchain & IPFS — [Springer Nature](#)
4. Blockchain-Enhanced Academic Certificate Verification: A Decentralized and Trustworthy Framework- [IEEE Xplore](#)
5. Design and Development of a Blockchain and IPFS-Based Credential Verifier with Custom Lightweight Encryption for Secure Storage – [IEEE Xplore](#)
6. Voice Controlled Home Automation with Cloud-Based Environment Monitoring System – [Springer Nature Singapore](#)
7. Certiblock: the exemplary utilization of blockchain for the rigorous validation of academic certificates – [IEEE Xplore](#)

Achievements

- **Champions**, SEED Global Hackathon, George Washington University & Boston University, 2024
- **Gold Medal**, International Research Day, SRM Institute of Science Technology, 2025
- **Champions**, Cybersecurity Ideathon, Department of NWC, SRM Institute of Science and Technology, 2026
- **Champions**, CTF, W3-CS, SRM Institute of Science and Technology, 2026
- **Champions**, Security Infrastructure Design, W3-CS, SRM Institute of Science and Technology, 2026
- **Gold Medal**, International Research Day, SRM Institute of Science Technology
- **Best Oral Presentation Award**, International Conference on Electronics and Informatics, 2024
- **Champions**, Pitching Competition, International Professional Activities Conference 2023, IEEE
- **Vice-Chancellor's Honors Scholarship**, ULAB, Summer 2021
- **Deans Honors Scholarship Awards**, ULAB, Fall 2022
- **Founder's Scholarship**, SRM Institute of Science and Technology, 2024
- Runners Up, **National Blockchain Olympiad Bangladesh**, 2023
- **Finalist**, CTOforum Hackathon
- 1st runners up, GP Academy Cisco Cybersecurity quiz

Security Research and Vulnerability Disclosure

- Discovered OTP Authentication Bypass vulnerability affecting Samsung India account registration workflow through HTTP response manipulation and session validation weaknesses.
- Identified OTP Verification Bypass vulnerability in Frido authentication flow that could allow unauthorized access.
- Conducted web application security testing using Burp Suite, proxy interception, session analysis, and authentication assessment.

Portfolio Website

itsraatul.github.io